**UNIVERSITY OF MACAU**

Faculty of Science and Technology

## CISC7204 Data Science and Data Visualization

Assignment 02 · Revised Project Proposal (Pair Work)

# Post-Pandemic Recovery of Macau's Visitor Economy: How Fast, and How Uneven?

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Pair 04 · Class D | Coursework item: Assignment 02 – Revised Project Proposal

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## 1. Project Title and Research Question

**Title**

### Post-Pandemic Recovery of Macau's Visitor Economy: How Fast, and How Uneven?

**Research question**

*Since Macau reopened its borders in January 2023, how has the monthly visitor volume recovered relative to the 2019 pre-pandemic baseline, and which visitor segments — same-day versus overnight, source market, and point of entry — have recovered fastest or lagged behind?*

**Region and domain**

Region: the Macao Special Administrative Region (SAR), China. Domain category: tourism and hospitality economics — specifically inbound visitor demand and the structure of the visitor mix.

## 2. Purpose

**Outcomes**

The project connects directly to the course outcomes on visual storytelling: it takes a real, government-published dataset and turns it into a series of charts that answer one question. It also supports our major in data science, where the ability to communicate a time-series result to a non-technical audience — policy makers, hotel operators, airline planners — is the deliverable that matters most. For Liu Haixiang it extends the recession-analysis work of the individual part of this assignment from a firm-level dataset to a whole destination economy; for Chan Ka Wai it is a first end-to-end project in which the statistics, the visual design and the narrative must agree with each other.

**Skills**

Data wrangling of official statistical releases (wide-to-long reshaping, month parsing, handling revised and provisional figures); index construction against a pre-pandemic baseline; time-series and composition visualization (index lines, small multiples, 100% stacked areas); interactive dashboard design in Plotly and Dash; and concise analytical writing that states what a chart does and does not show.

**Knowledge**

Tourism demand and seasonality; structural breaks and level shifts in time series; the difference between volume recovery and value or quality recovery; the composition effects created by same-day visitors; and the limits of official monthly statistics when definitions change.

## 3. Data Sources

Both datasets are published by the Statistics and Census Service (DSEC) of the Macao SAR and are free to download. They cover the same population and the same monthly frequency, so they can be joined on the month key.

| # | Dataset and publisher | Link |
|---|---|---|
| **1** | Visitor Arrivals — monthly, 2019 to present (DSEC, Tourism Statistics) | https://www.dsec.gov.mo/en-US/Statistic?id=402 |
| **2** | Package Tours and Hotel Occupancy Rate — monthly, 2019 to present (DSEC) | https://www.dsec.gov.mo/en-US/Statistic?id=403 |
| **3** | Visitor Statistics Database — detailed, queryable and exportable segment data (DSEC) | https://www.dsec.gov.mo/TourismDBWeb |

Data volume is small (about 90 monthly observations per series), which is an advantage: every number can be traced back to a published table, and the whole analysis runs in a single notebook.

## 4. Tasks and Deliverables

- **Task 1 — Acquisition.** Download the monthly visitor-arrival tables and the package-tour / hotel-occupancy tables; record the download date and the release status of the last month in a data dictionary.
- **Task 2 — Tidying.** Reshape every table into a long format with the keys month, year, series, value and unit; parse the month labels into real dates; keep the original source table name as a column so any figure can be audited.
- **Task 3 — Baseline index.** Set 2019 = 100 for total arrivals and for each segment (same-day, overnight, by source market, by point of entry) so that series of very different magnitudes can be compared on one axis.
- **Task 4 — Visual analysis.** Produce (a) an index line chart of total monthly arrivals with the border-reopening date marked; (b) a year-by-month heat map to expose seasonality; (c) small multiples of the index by source market; (d) a 100% stacked area of same-day versus overnight visitors; (e) a hotel-occupancy line to compare volume with capacity utilisation.
- **Task 5 — Dashboard.** Build a Plotly Dash dashboard with two dropdowns (report type and month or year) that reproduces the charts interactively, reusing the component pattern practised in Module 04.
- **Task 6 — Storytelling.** Write the analytical narrative — five to six slides, one message per slide, each slide stating the question, the chart and the one-sentence conclusion.

**Deliverables**

- cleaned long-format dataset (CSV) plus data dictionary;
- Jupyter notebook that reproduces every figure from the raw download;
- Dash application (zipped project space) with the two-dropdown layout;
- five-slide visual story in PDF, and a two-page written summary of findings.

## 5. Criteria for Success

A finished product is successful when a reader who has never seen the data can, within one minute, state what happened to visitor volume, when the turning point was, and which segment recovered slowest — and can then check that claim against a labelled chart. Concretely:

- every figure is reproducible from the downloaded files with one command;
- each chart carries a title that states the message, labelled axes with units, and a readable source note;
- no chart mixes two y-axes or compares index levels that are not on a common baseline;
- the dashboard runs locally from the zipped project space without editing the code;
- the written summary contains no claim that cannot be pointed to on a chart.

**Assessment rubric (self-check before submission)**

| Criterion | Weight | Meets the standard when… | Falls short when… |
|---|---|---|---|
| **Correctness of analysis** | 30% | index and segment figures match the published tables; assumptions stated | numbers cannot be traced to a source table |
| **Choice of visual form** | 25% | each chart type fits the question (trend, composition, comparison) | charts chosen for decoration; unreadable defaults |
| **Clarity of storytelling** | 25% | one message per chart; titles are conclusions, not labels | reader must be told what to see |
| **Reproducibility** | 20% | notebook and dashboard rerun end-to-end from raw data | manual edits or missing files |

## 6. How the Visualization Answers the Question

The research question has two halves, speed and unevenness, and each half needs a different visual form. Speed is a question about a single series over time, so it is answered by an index line chart with 2019 set to 100: read vertically, the line shows how many months it took to return to the pre-pandemic level, and read horizontally it shows whether the recovery stalled in particular seasons. Unevenness is a question about composition, so it is answered by small multiples of the same index for each source market and by a 100% stacked area of same-day versus overnight visitors — the first shows which markets are carrying the recovery, the second shows whether the recovery is made of high-value overnight stays or of day-trippers who spend little. Pairing the visitor series with the hotel-occupancy series adds the capacity dimension: arrivals can recover while occupancy does not, and that gap is itself a finding.

Because both datasets are monthly and official, the dashboard can let a reader switch between markets and months and see the same story from a different angle, which is exactly the ‘storytelling with data’ behaviour the course asks for. The visualization therefore does not merely display the numbers: it answers the two-part question by construction, one chart per part, and it keeps a visible baseline so that no reader can mistake a seasonal dip for a new shock.

## 7. Collaboration

Division of work. Chan Ka Wai owns the data pipeline (Tasks 1–3): downloading, tidying, the data dictionary and the baseline index. Liu Haixiang owns the visual layer (Tasks 4–5): the chart specifications, the Dash dashboard and its two dropdowns. Task 6, the story, is written jointly, with each member drafting the slides for the charts they built and the other reviewing them. Both members review the final notebook line by line before submission.

**Ground rules**

- one shared folder per member, one shared notebook, merged weekly — no separate copies of the notebook;
- every figure in the story must be reproducible by the other member on their own machine;
- no number enters a slide unless it can be traced to a DSEC table and a date;
- any use of an AI assistant is recorded in the notebook (prompt, tool, what was accepted), following the course declaration practice;
- a 15-minute check-in at the start of each work session, and a five-day buffer before the deadline.

**Roles**

| Role | Member and responsibility |
|---|---|
| Project coordinator | Liu Haixiang — schedules the check-ins and owns the final submission checklist. |
| Project scribe | Chan Ka Wai — keeps the decision log and the data dictionary. |
| Progress tracker | Chan Ka Wai — maintains the task board and flags slips more than two days. |
| Liaison | Liu Haixiang — communicates with the instructor and the TA on scope questions. |